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Convolutional neural network for high-performance reservoir computing using dynamic memristors

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ABSTRACT

In the rapidly advancing field of neuromorphic computing, W/ZnO/TiN resistive random-access memory (RRAM) devices have emerged as a next-generation computational building block. Our findings reveal the significant role played by the thickness of the ZnO layer in determining the electrical properties essential for data storage and neuromorphic applications. The short-term memory (STM) capabilities, which are critical for processing temporal information, are closely examined alongside their potential to simulate biological synaptic functions through multilevel conductance states and synaptic behaviors such as paired-pulse facilitation. Integrating these devices into reservoir computing systems enhances pattern recognition and accelerates learning, which demonstrates their utility in sequential data processing. In addition, conductance modulation via pulse width adjustment is a novel strategy to optimize memory device performance. By showcasing the effectiveness of W/ZnO/TiN devices in neuromorphic computing through high-accuracy image recognition tasks, our study highlights their foundational role in advancing neuromorphic computing technologies. The adaptability, learning capabilities, and efficiency of these devices underscore their potential for developing hardware-based neuromorphic systems that are capable of complex data processing.

1. Introduction

The rapid advancement of computing technology has substantially impacted our lives and made ongoing improvements in data processing and storage technologies essential. The "von Neumann architecture", once the cornerstone of computing, now struggles to process the large amounts of data required by modern applications such as Artificial Intelligence (AI) and the Internet of Things (IoT). Recently, neuromorphic computing, which mimics the human brain, has emerged as a potential solution to these challenges [1–3]. Neuromorphic computing presents a promising alternative to energy-efficient computing and is anticipated to eventually replace the traditional von Neumann architecture completely. This computation type emulates biological synapses and neural systems, which makes it possible to process large-scale data while ensuring minimal power consumption [4]. The magnitude, retention duration, and repetition frequency of external signals and stimuli play crucial roles in processing and memory capabilities [5,6]. Next-generation memory devices have been developed to mimic the efficient structure of the human brain. Notably, various non-volatile memory (NVM) technologies are also explored, including ferroelectric random-access memory (FRAM) [7,8], spin-transfer torque random-access memory (STT-RAM) [9], phase-change random-access memory (PcRAM) [10,11], magnetic random-access memory (MRAM) [12], and resistive random-access memory (RRAM) [13–15]. Each technology offers unique advantages for integration into neuromorphic computing systems. Among these, RRAM stands out due to its reliable data storage, high integration density, low power consumption, and rapid processing speeds, establishing it as a critical element in advancing neuromorphic computing [16–19]. Therefore, this study focuses on resistive random-access memory (RRAM) as the core component of conducting filaments, which presents challenges for controlling multilevel switching operations due to stochastic variations and abrupt switching. This represents a significant problem when online training is implemented [20]. On the other hand, non-filamentary RRAM does not rely on the formation of conductive filaments but rather on mechanisms that change their resistive state through electrochemical doping or ion migration. Hence, this second type offers notable advantages for mimicking biological synapses in information processing [21].

This study focuses on W/ZnO/TiN-based RRAM with a non-filamentary switching mechanism that offer more reliable and

전도성 필라멘트를 핵심 요소로 사용하는 RRAM 확률적 변동성과 급격한 스위칭 특성 때문에 다중 레벨 스위칭을 제어하는 데 어려움이 있다.
필라멘트형 = 아주 가는 전도 경로(ON/OFF ration가 크다)
but 랜덤성이 크다(필라멘트가 매번 다른 위치), abrupt switching(전압을 조금 더 줬더니 저항이 확 바뀜)
비필라멘트형 RRAM은 전기화학적 도핑이나 이온 이동을 통해 저항 상태를 변화시킨다. = 통로가 소자 전체 또는 넓은 영역

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다중레벨 전도도(MLC) : 한 소자에 더 많은 정보 저장이 가능하므로 신경망의 가중치(weight) 역할,
전압 펄스에 따라 conductance가 변하므로 reservoir 내부 상태 표현력 증가-> classification 성능 향상

쌍자극 촉진(PPF) : 두 개의 자극을 짧은 간격으로 주면 두 번째 반응이 더 커지는 현상(short-term memory 특성)

predictable performance. This characteristic is crucial for developing memory systems that not only store data but also emulate the dynamic essence of human memory, ranging from short-term memory (STM) to long-term memory (LTM). At the heart of our exploration is the capability of the device to simulate the complex learning and recognition processes of neural networks. With features such as multilevel conductance (MLC) and paired-pulse facilitation (PPF), W/ZnO/TiN RRAM offers an essential component for advanced data processing that utilizes high-dimensional dynamic states. Moreover, this device enhances classification recognition and data processing abilities by utilizing dynamic states that are essential for reservoir computing (RC). The integration with advanced neural network architectures, especially convolutional neural networks (CNNs), significantly enhances the effectiveness of the RC system [22,23]. CNNs, which are renowned for their capabilities in image recognition and classification capabilities, improve the system by efficiently extracting and processing local features from input data. This synergy enables the RC system to perform complex pattern-recognition tasks, such as handwriting recognition, using datasets like MNIST. Incorporating W/ZnO/TiN RRAM devices into an RC framework signifies a significant step toward realizing the full potential of neuromorphic computing. This study highlights technological progress and demonstrates the transformative impact these innovations will have on future applications across computing paradigms.

dynamic state: 입력을 받으면서 시간에 따라 계속 변하고, 이전 입력의 흔적도 일부 남아 있는 내부 상태

2. Experimental

The W/ZnO/TiN device was fabricated using the following process: First, a 100 nm thick TiN bottom electrode was deposited on a SiO_2/Si wafer using radio frequency (RF) sputtering. A ZnO film, with an approximate thickness of 40 nm, was fabricated via RF sputtering. This process utilized 80 W of power and involved the use of Ar gas (8 sccm) and O_2 (12 sccm) gases. The sputtering process was specifically targeted toward the deposition of zinc. Subsequently, a circular top tungsten (W) electrode was deposited with a thickness of 50 nm via direct current (DC) sputtering. This deposition was conducted under an operational pressure of 3 mTorr. The electrical characteristics of the device were measured using a Keithley 4200-SCS semiconductor parameter analyzer (SPA) and a 4225-PMU pulse measurement unit from the probe station. In addition, a bias was applied to the W top electrode while the bottom electrode was grounded.

3. Results and discussion

3.1. Structural element analysis of the RRAM device

ZnO, TiN, TiON의 성분 조사

Each layer's existence and distinguishable boundaries were validated through transmission electron microscopy (TEM) analysis, as illustrated in Fig. 1(a). A schematic of the fabricated devices is shown in Fig. 1(b). By moving the probe to contact the top electrode, voltages were applied while the bottom electrode was grounded. EDS analysis was conducted to consider the impact of composition on the measured data. Moreover, the atomic concentration as a function of depth is displayed in Fig. 1(c). The concentrations of the Zn atoms dropped, while the O atom concentration increased in the storage layer between 85 and 95 nm as the depth approached the TiN electrode. Furthermore, Ti atoms entered the switching layer with a deeper penetration than the N atoms. This finding directly confirms the presence of a thin TiO_x interfacial layer. Since TiO_x has a Gibbs free energy of −888.9 kJ/mol, it is substantially lower than that of ZnO (−350 kJ/mol) and TiN (−308/9 kJ/mol). As a result, TiO_x or TiON can be formed by collecting O^{2-} ions, which act as an oxygen reservoir [24].

Additional research on the ZnO-formed structures using XPS analysis reveals the distinctive Zn 2p doublet peaks in Fig. 2(a), which correlate with the Zn^{2+} oxidation state [25,26]. Zn $2p_{3/2}$ and $2p_{1/2}$ are represented by the components centered at binding energies of 1021.8 eV and 1044.9 eV, respectively. The Zn $2p_{3/2}$ and $2p_{1/2}$ core level binding energies slightly differ from those reported in other studies [25–27]; however, their separation remains consistent at 23 eV. There is also an asymmetric peak in the O 1s area of the XPS spectrum, which suggests the presence of several oxygen species. As depicted in Fig. 2(b), the curve was fitted to four distinct Gaussian-Lorentzian GL components: peaks 1, 2, and 3. Peak 2, observed at 532.8 eV, is associated with the O^{2-} state of oxygen defects or vacancies, which contribute to the formation of the non-stoichiometric ZnO. Conversely, peak 1, situated at 532.2 eV, is attributed to O^{2-} ions involved in the Zn—O bonding within the wurtzite structure of Zn^{2+} [25,28,29]. Lastly, like OH groups, weak oxygen bonds on the surface are usually associated with peak 3 (at 534.4 eV). Fig. 2(c) presents the XPS spectrum of N 1s. The primary peak is situated at 400.3 eV, which corresponds to TiN, while the secondary peak (at 401.6 eV) is attributed to the TiON layer, indicating surface oxidation of the TiN bottom electrode [30]. Fig. 2(d) shows the XPS spectrum and Gaussian fitting of multiple peaks for Ti 2p core-level electrons to monitor the chemical information of the interface layer

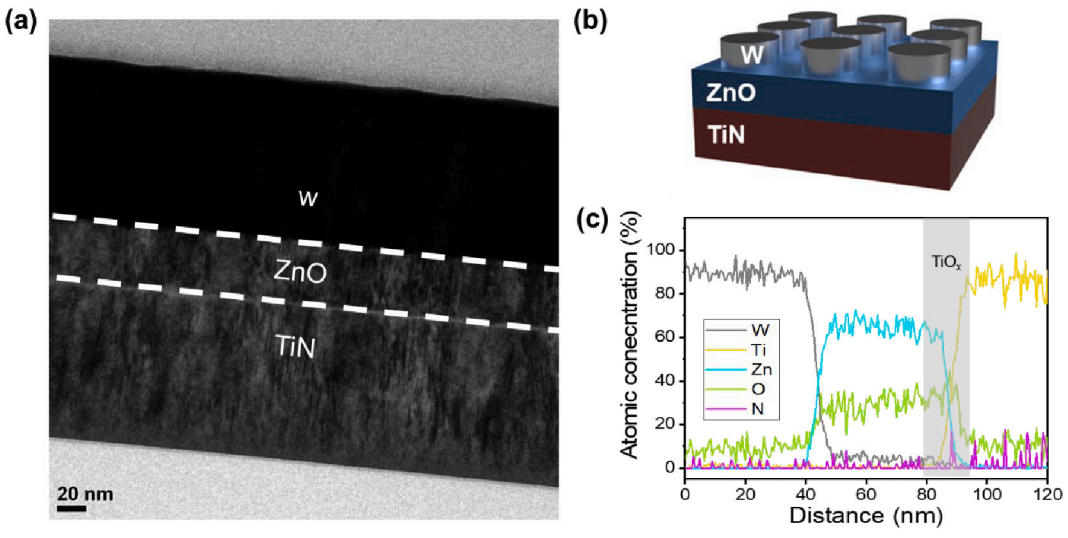
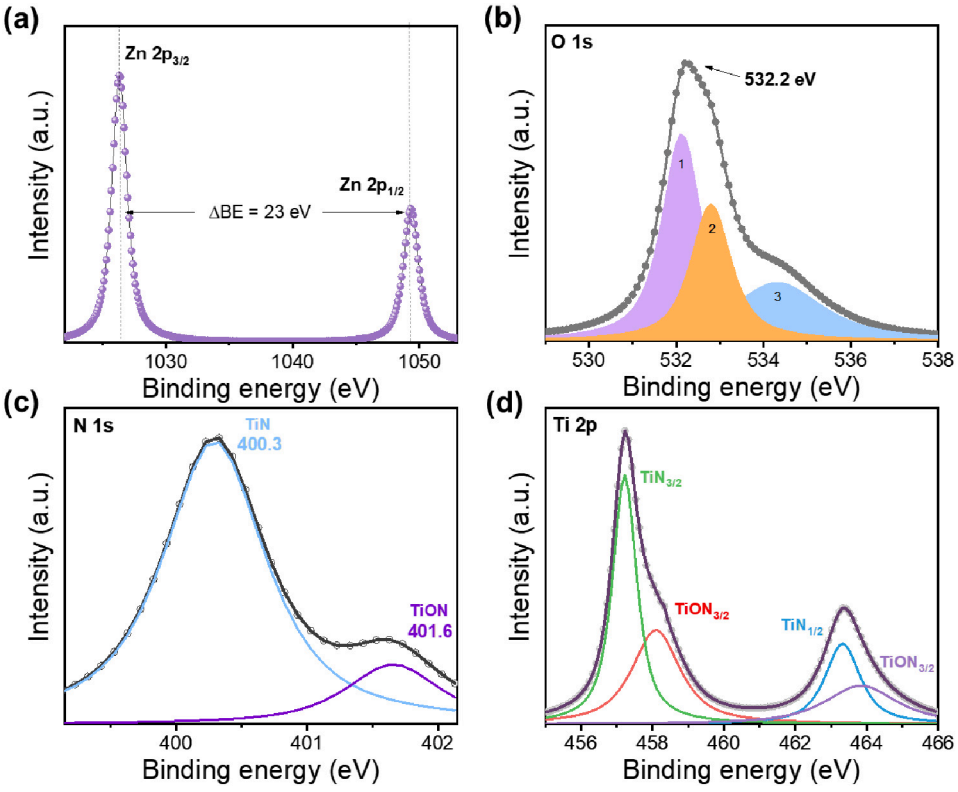


Fig. 1. (a) Cross-sectional image recorded with a TEM. (b) Schematic image of the device when measuring electrical properties with a Keithley-4200 SCS. (c) EDS weight percentage composition.

(a) 이미 Zn2+상태인 ZnO에 고정된 에너지인 x-ray를 쏴주면
전자를 잃은 상태로 binding energy shift된 피크 위치를 보고 Zn2+인 것을 확인
n=2, l=1, s=+-1/2인 두개의 binding energy. = doublet

(b) GL components(비대칭) -> peak1 : 정상격자산소(Zn2+ - O2- - Zn2+) = wurtzite structure, = bulk(TiN과 가까운 부분)
peak2 : 결함 주변 산소(Zn - □ - Zn) = oxygen vacancy(defect) = non-stoichiometric ZnO,
peak3 : 표면 흡착 산소 OH(screening감소, 핵 당김 증가) => binding energy shift

TiN의 표면이 산화되어 TiON의 얇은 계면층이 생성
=> 저항 스위칭 역할
TiN = 이온성 + 공유성 + 금속성
Ti-O-N > Ti-N (결합에너지)



TiON1/2인데 잘못 적힌건가

Fig. 2. XPS spectra of (a) Zn 2p, (b) O 1s, (c) N 1s, and (d) Ti 2p for the W/ZnO/TiN stack.

between ZnO and TiN. Furthermore, peaks indicative of Ti—N and Ti-O-N bonds were observed, which suggests the formation of a TiON interface layer that plays a crucial role in resistive switching [31]. The peak for Ti-O-N bonds is located at higher binding energies, specifically 458.31 eV and 463.43 eV, compared to the Ti—N bonds. This is consistent with a distinct chemical environment within the interface layer.

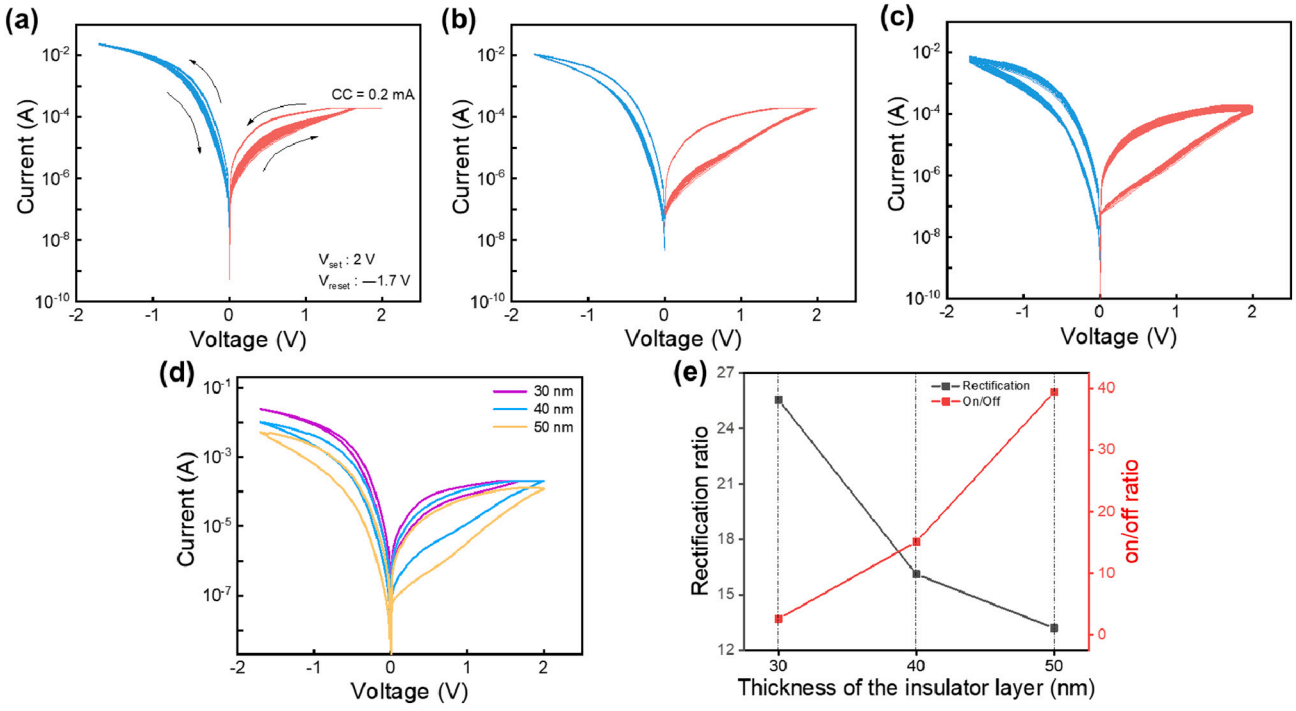


Fig. 3. Despite identical set and reset voltage (2 V, −1.7 V), the shape of the IV curve varies depending on the insulator layer's thickness. Thickness of the used ZnO: (a) 30 nm, (b) 40 nm, (c) 50 nm. (d) IV curves for each layer thickness. (e) Comparison of on/off and rectification ratios across thicknesses.

forming process
처음에 강한 전압을 걸어서 절연체 내부에 필라멘트를 생성
non-filamentary RRAM에서는 defect-assisted conduction network
마지막 문장 해석 = 모든 소자가 별도의 forming 과정 없이 스위칭을 보였으며, 이는 ZnO 내부의 풍부한 결함(특히 oxygen vacancy)이 전하 트랩 및 전도 경로 역할(hopping->메모리 효과)을 하기 때문이라고 해석된다.

3.2. DC measurements

The electrical characteristics of W/ZnO/TiN resistive random-access memory (RRAM) devices were investigated by varying the thickness of the ZnO insulating layer (30 nm, 40 nm, 50 nm). The analysis was conducted through the observation of the I-V (current-voltage) properties of the devices, following the application of a direct current (DC) sweep voltage ($0 \rightarrow +2 \rightarrow 0 \rightarrow -1.7 \rightarrow 0$ V) to the top electrode, using 100 consecutive switching cycles. These measurements were obtained under a compliance current of 200 μA, as depicted in Fig. 3(a)-(d). All ZnO-based RRAM devices (thicknesses: 30, 40, and 50 nm) exhibit characteristic behavior without requiring an electrical forming process. This behavior can be attributed to the presence of numerous defects within the device insulator, which serve as charge traps.

양전압 = ZnO 내부 defect 재배열 (특히 산소 이온 이동, Oxygen Vacancy network 강화)

Furthermore, anodic bias induces the set process, while cathodic bias triggers the reset process. The initial state of the device shifts to a low resistance state (LRS) via a set process induced by an anodic bias DC sweep. This promotes the growth of conductive defects, which facilitates the transition to a low-resistance state. The reset process, which is induced by a cathodic bias DC sweep, reduces conductive defects and shifts the state to a high resistance state (HRS). All devices show gradual resistive switching characteristics, and a self-rectifying effect is observed. Fig. 3(e) shows the rectification ratio and the on/off ratio corresponding to each thickness of the ZnO layer. It was also observed that the on/off ratio of RRAM devices increased with the thickness of the ZnO layer, which suggests that an increase in the on/off ratio enhances not only device reliability but also the performance for data storage and retrieval.

자기 정류 기능
+V 전류 큼
-V 전류 작음

Conversely, the rectification ratio demonstrated a decreasing trend as the thickness of the ZnO insulating layer increased. This reduction could potentially affect the efficiency of devices in applications where a non-linear current response is crucial. Notably, the devices with a 40 nm thick ZnO layer showed a balanced trade-off between the on/off ratio and the rectification ratio. This equilibrium is critical for applications in synaptic devices, which require effective signal differentiation through a high on/off ratio and non-linearity modulation by choosing a suitable rectification ratio. All 40 nm thick devices satisfy these requirements and are thus reasonably optimized for synaptic measurements and applications such as neuromorphic computing and biological neural networks.

Fig. 4(a) demonstrates a decrease in current values over time upon applying a read voltage, thereby verifying the W/ZnO (40 nm)/TiN device as a switching Tunneling Memory (STM). The device shifts to an LRS when a set voltage of 2.5 V is applied, with a compliance current (CC) of 200 μA. After a 5 s delay, applying a 0.5 V read voltage reduces the current significantly. This indicates that the LRS is not sustained, i.e., that the W/ZnO/TiN device exhibits volatile characteristics that can lead to data loss over time. Fig. 4(b) shows a box plot of the HRS and LRS resistance values measured at 0.5 V for ten cycles of ten devices. All devices demonstrate a memory window greater than 5, with no abrupt changes between HRS and LRS, which confirms the stability of the devices across multiple devices and cycles. Fig. 4(c) displays the resistance numbers for the LRS and HRS over 500 DC switching cycles, which characterize the device's durability. The analysis revealed an ON/OFF ratio of approximately 3.39, demonstrating that the device consistently exhibits a high capacity for information differentiation, a key indicator of its read and write endurance. Both the consistent resistance values and ON/OFF ratio throughout the 500 cycles demonstrate the ability of RRAM devices to deliver stable performance over numerous cycles. Fig. 4(d) shows the rate of change in conductance for LRS and HRS for 10,000 s during a retention measurement. As time progresses, following the set and reset processes, the resistance values of both LRS and HRS fluctuate. These fluctuations involve rapid increases or decreases before converging to a stable point. The continuous resistance changes in LRS

<구조>
W(top/+)
ZnO
TiON/TiN(bottom/-)
-> 양전압 기준

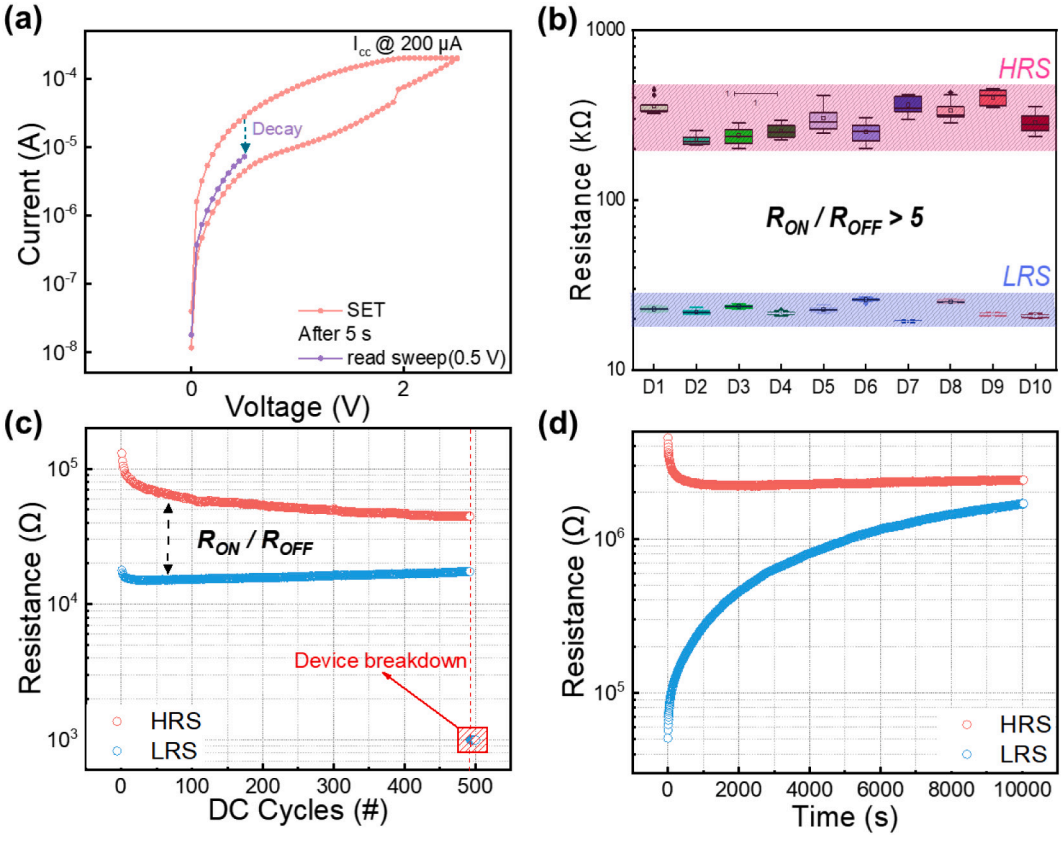


Fig. 4. (a) DC read sweeps following the set process with a CC of 200 μA; the inset shows the magnified graph from 0 V to 0.5 V. (b) Device-to-device uniformity. (c) Endurance characteristics at a read voltage of −0.2 V. (d) Retention characteristics for a read voltage of −0.2 V.

and HRS underline the volatile nature of the device, which emphasizes its potential for STM applications. The applicability of short-term plasticity (STP) in processing and computing information presents several advantages in neuromorphic computing. Its low-power operation and high speed enable rapid data processing while minimizing unnecessary storage requirements. Furthermore, STP is a robust foundation for the non-electroforming progressive transition model, which supports the volatility inherent in the STM transition mechanism [32].

3.3. Switching mechanism of W/ZnO/TiN device

We investigated the switching mechanism of the W/ZnO/TiN device and focused on the crucial role of TiON layer formation and modulation. Unlike the through-filament switching models, our device demonstrates non-filamentary switching characteristics governed by the migration of oxygen ions and vacancies, as well as the modulation of the interfacial TiON layer thickness [33–35]. Fig. 5(a) shows the device in its initial state. Initially, due to the lower standard Gibbs free energy for the formation of TiON compared to ZnO, an intrinsic TiON layer formed within the W/ZnO/TiN structure [36]. The interfacial layer naturally modulates the resistance state of the device, which underscores a forming-free characteristic that reduces the energy requirements for switching. This enhances both the stability and energy efficiency of the device. The resistive switching in this non-filamentary RRAM device is attributed to a barrier modulation mechanism facilitated by the migration of oxygen ions.

Upon applying an electric field, oxygen ions migrate within the ZnO layer, which dynamically modulates the thickness of the TiON interfacial layer. This process is expedited by the intrinsic energy characteristics of the materials involved and oxygen migration being strongly affected by the direction and strength of the electric field. Fig. 5(b) shows the process where oxygen ions migrate through the ZnO layer, which decreases the thickness of the TiON layer (LRS). In addition, reducing the thickness increases conductance at the ZnO/TiN interface, which decreases device resistance. Conversely, Fig. 5(c) demonstrates that oxygen ions move back toward the ZnO/TiN interface during reset, which thickens the TiON layer. This expansion inhibits electron movement and causes the device to switch back to HRS. This also indicates that the conductance change in the W/ZnO/TiN device is a gradual process that relies on the movement of oxygen ions and the adjustment of the TiON layer thickness. In conjunction with this delicate modulation process, the inherent forming-free characteristic of the device highlights its potential for reliable, low-power, and efficient memory applications. Moreover, the consistent resistance states and ON/OFF ratios (maintained across numerous switching cycles) confirm that the device can perform reliably over extended periods, i.e., it is suitable for non-volatile memory applications.

3.4. Multilevel conductance characteristic of the W/ZnO/TiN device

The investigation of the multilevel conductance (MLC) characteristics was conducted on W/ZnO/TiN devices during DC sweeping by adjusting the current compliance (I_{CC}) and the reset voltage (V_{RESET}). Importantly, this approach reveals the ability to progressively adjust resistance values via the MLC characteristics, which suggests the possibility of mimicking the operational mechanisms of biological synapses. This extends beyond the traditional functionalities of memory devices and helps explore novel methods for processing and storing information (e.g., biological neurons). Fig. 6(a) shows the MLC characteristics where the I_{cc} varies from 100 μA to 1 mA at a read voltage of 1 V. As I_{cc} increases, the movement and diffusion of oxygen vacancies increase, which expands the conductive channels and increases the current flow in the LRS. Under these conditions, the observed resistance values gradually decrease from 10.1 kΩ to 2.16 kΩ. This represents an overall decrease of approximately 78.6 %. Fig. 6(b) shows the characteristics of the MLC as a function of V_{RESET} changes. The observed resistance values increase as the magnitude of V_{RESET} gradually increases from −0.8 V to −2.2 V. This indicates that at the highest V_{RESET}, the resistance increases by approximately 3.31 times compared to its initial value. These results indicate that an increase in V_{RESET} can significantly enhance the rearrangement of oxygen ions and restrict conductive channels, which leads to a uniform change in the HRS and demonstrates a stable distribution of oxygen vacancies within the switching layer [37–40]. The implementation of this MLC functionality underscores the possible application of W/ZnO/TiN devices as synaptic memristors, which are essential for mimicking complex information processing and learning mechanisms in biological networks.

3.5. Application as a synaptic memristor

Fig. 7(a) illustrates the potentiation and depression curves of a synaptic device when a read voltage of 0.2 V is applied. The set pulse amplitude with a pulse width of 50 ms is 1.5 V, while the magnitude of the reset pulse is −1.5 V. Fig. 7(b) shows the conductance changes in a W/ZnO/TiN device after exposure to 50 repeated P&D set/reset pulses with amplitudes of 2 V/−1.5 V, for a range of pulse widths (5, 10, 20, 30, 40 ms). This study reveals significant and progressive changes in conductance for each pulse width, which indicates a sensitive response of the device to pulse-width alterations. Notably, an expansion in the dynamic conductance range is observed as the pulse width increases. This finding highlights the ability to finely control the conductance changes induced by a fixed pulse amplitude by adjusting the pulse width. This insight highlights the potential for optimizing memory device processes by precisely modulating the electrical input parameters. Fig. 7(c) demonstrates the transition from short-term plasticity (STP) to long-term potentiation (LTP) upon applying solely a read voltage after

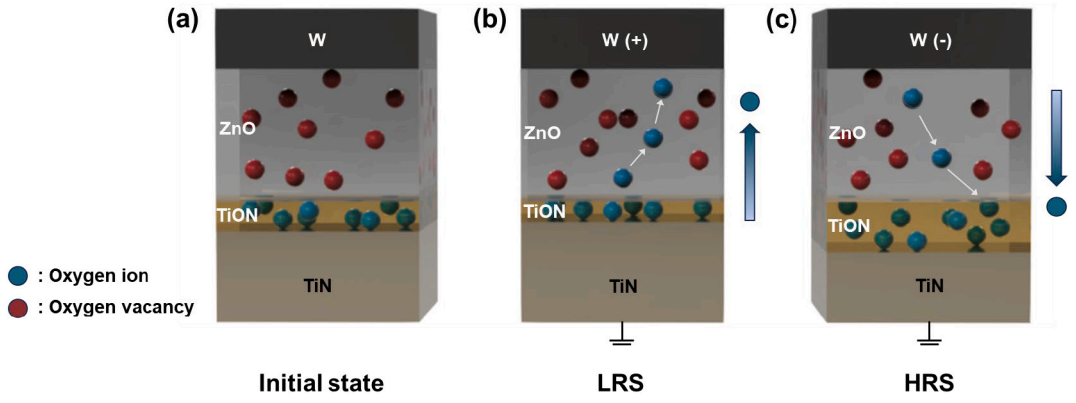


Fig. 5. Schematics illustrating the RS mechanism for (a) the initial state, (b) LRS, and (c) HRS.

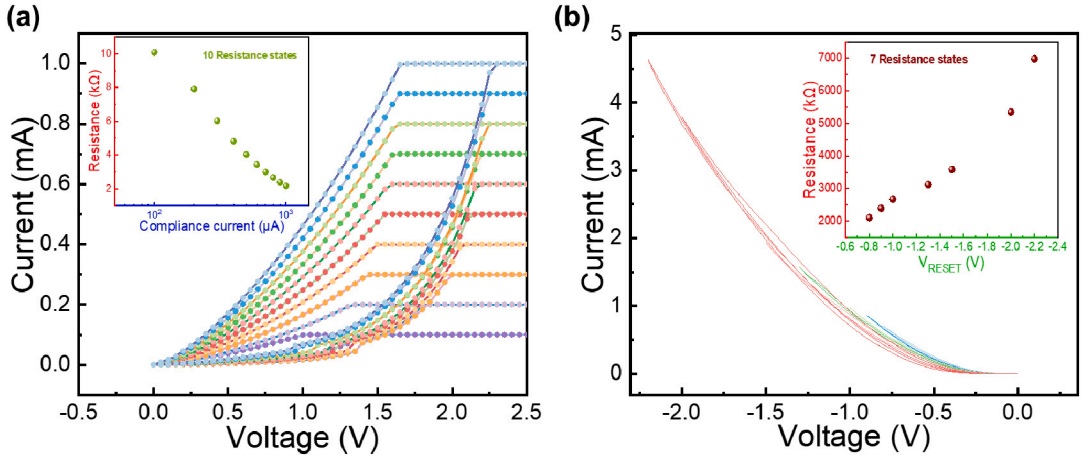


Fig. 6. Multilevel variations based on CC values for a read voltage of 1 V. (a) LRS, (b) HRS.

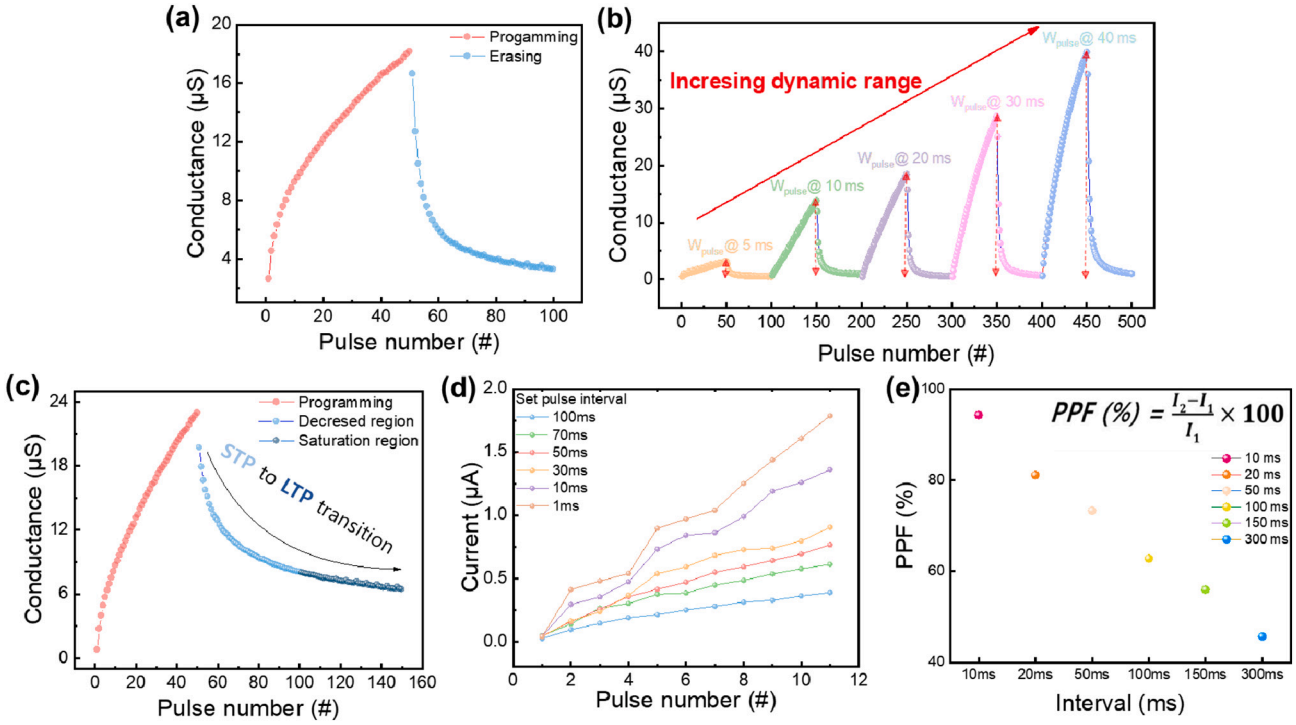


Fig. 7. (a) Potentiation and depression linearly measured using incremental step pulses. The pulses used for potentiation and depression have a width of 10. (b) Synaptic plasticity of memristors exhibiting potentiation and depression for different pulse widths. (c) Only a read operation at 0.2 V was performed following potentiation. (d) Conductance as a function of pulse number with different pulse intervals for potentiation. (e) PPF index as a function of interval time.

potentiation (without any subsequent reset pulses). After a series of potentiation processes, a read voltage of 0.2 V was applied to evaluate the device's conductance stability and memory retention capability. This method confirms the ability of synaptic devices to transition from a state of temporary plasticity (where changes are rapidly reversible) to a more durable state of potentiation (indicative of long-term memory formation). Fig. 7(d) illustrates the changes in conductance as a function of pulse number, depending on the pulse interval. The applied set voltage is 2 V, and the read voltage is 0.2 V. As the number of pulses increases, the conductance increases gradually.

Furthermore, the narrower the interval between pulses, the faster the increase in conductance. Fig. S1 (Supporting Information) shows the current variation for different pulse intervals. The application of set pulses leads to a progressive increase in current, whereas in the absence of set pulses, the current decreases gradually.

Additionally, we investigated the effects of paired-pulse facilitation (PPF) in the W/ZnO/TiN RRAM device. This phenomenon is widely known in biological synapses for playing a crucial role in short-term synaptic plasticity and temporal information processing [41,42]. An experiment was conducted to demonstrate the STM effect of the device by employing two identical pulses (amplitude: 2.5 V, duration: 10 ms) separated by different time intervals. The intervals were set to 10 ms, 20 ms, 50 ms, 100 ms, 150 ms, and 300 ms. The intervals between pulses were set to 10 ms, 20 ms, 50 ms, 100 ms, 150 ms, and 300 ms. Fig. 7(e) shows the PPF index within a single cell, which indicates the rate of change between the first and second pulses. The PPF was calculated using the following formula:

$$\mathrm{PPF} = \frac{I_2 - I_1}{I_1} \times 100\ (\%) \tag{1}$$

Here, I_1 is the average current of the first pulse, and I_2 denotes the average current of the second pulse [43]. The PPF values for each interval were 94.28 %, 81.12 %, 73.31 %, 62.82 %, 56.01 %, and 45.73 %, respectively. These numbers demonstrate the capability to modulate synaptic weight by altering the intervals between consecutive pulse stimulations. Notably, higher PPF values at shorter intervals (e.g., 10 ms) indicate enhanced responsiveness to rapid consecutive stimulations, whereas decreasing PPF values at longer intervals (e.g., 300 ms) reflect a reduction in responsiveness. Such results suggest that the device can emulate short-term effects often observed in biological synapses, which highlights its potential to play a significant role in temporal information processing.

3.6. Reservoir computing system

Finally, Reservoir computing (RC) is a computational framework primarily focused on processing temporal and sequential data inputs in parallel to facilitate rapid learning. Fig. 8(a) consists of an RC schematic for mapping inputs to a high-dimensional space and a readout component for pattern analysis from the high-dimensional state of the reservoir. The MNIST database was used to verify the capabilities of the image recognition method. The learning process involved a readout layer based on a convolutional neural network (CNN), a type of artificial neural network specialized in recognizing and interpreting local patterns in images [44]. The reason for using a CNN as the readout layer is that it significantly reduces the number of parameters compared to fully connected neural networks (FCNNs) [45].

Moreover, to mitigate the computational load on the hardware of the neuromorphic system, our approach minimizes the number of synaptic elements by leveraging a CNN architecture with fewer parameters than reported conventional networks [46–48]. In order to assess the potential applicability of 4-bit reservoir computing, the standard MNIST data of 28×28 pixels were first binarized as a preprocessing step. Next, the data were grouped into sets of four pixels each, using 4-bit reservoir computing data. Consequently, a total of 7×28 groups were generated, each representing one of the various states from [0000] to [1111]. The processed image was entered into 196 virtual array memristors, which underwent high-dimensional mapping based on 4-bit reservoir computing data. The output from the reservoir layer was then fed into a CNN-based readout layer. This CNN model incorporates two convolution layers, an average-pooling layer, and several fully connected layers to optimize image pattern recognition. The initial convolution layer had a 3×3 kernel size, which accommodates an input channel of 1 and produces an output of 32 channels, with a stride of 1 to maintain spatial resolution. Subsequently, the second convolution layer expanded compared to the first by increasing the input channel count to 32 and the

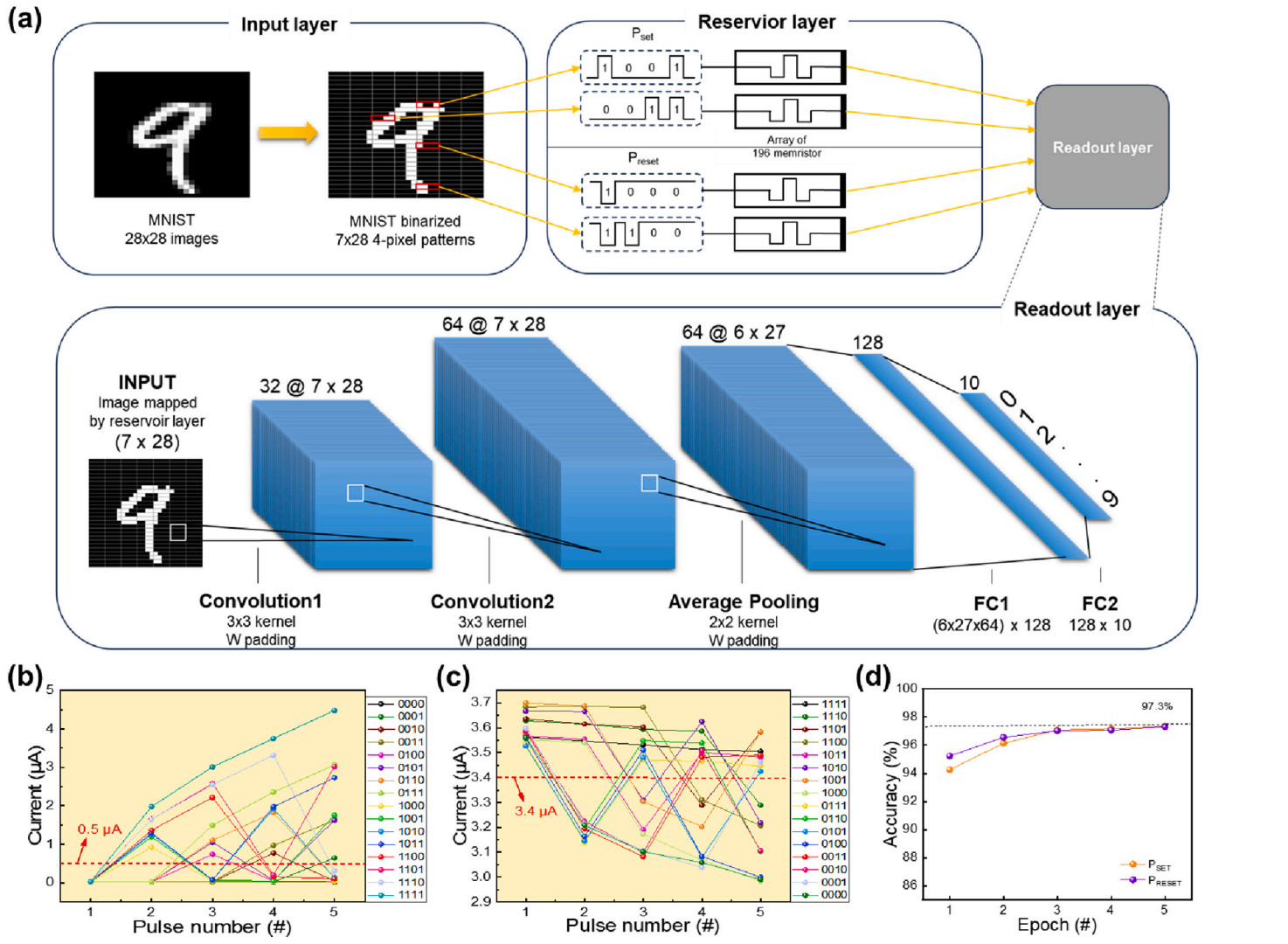


Fig. 8. (a) The concept of reservoir computing and the designed CNN architecture for MNIST dataset classification. 16 states configured by specific pulse sequences: (b) LRS mode, (c) HRS mode. (d) Dependency of pattern accuracy on the number of training epochs.

output to 64. This preserved both kernel size and stride, thus enhancing the network's ability to capture more complex features. To ensure no image data loss and maintain the original spatial dimensions, zero-padding was applied across all convolution layers. Combined with the ReLU activation function, this technique introduces non-linearity, which can help identify more nuanced patterns within the data. Batch normalization was also employed post-activation to accelerate the learning process by stabilizing the distribution of the inputs to subsequent layers. After the data progressed through the convolution layers, a dimensionality of $7 \times 27 \times 64$ was achieved. They then underwent further refinement through an average pooling layer, which reduced its size to $6 \times 27 \times 64$. This step is crucial for mitigating noise and compressing the image data without sacrificing critical information. The processed data were subsequently fed into a dense layer of 128 hidden neurons, establishing a deep learning framework that is conducive to advanced pattern recognition and classification tasks. This structure ensures that CNN is adept at processing and interpreting local patterns within images, efficient in training, and adaptable to various image-based applications. Fig. 8(b)-(c) displays a total of 16 states, ranging from [0000] to [1111], represented in a 4-bit format through a W/ZnO/TiN RRAM device exhibiting the STM characteristics.

In utilizing the RC leveraging positive pulse fading (PPF) characteristics, as depicted in Fig. 8(b), a 200 ms delay is incorporated along with a P_{SET} voltage of 2 V. Conversely, the RC exploiting paired-pulse depression (PPD) traits, shown in Fig. 8(c), used a 50 ms delay and a P_{RESET} voltage of −1.7 V to capture the memristor's reactions across 16 varied 4-bit pulse trains. In order to ascertain the initial condition of the memristor, the conductance reading for the first bit of each of the 16 sequences was acquired with a P_{read} voltage of 0.5 V. The conductance measurements for the data encoded in the subsequent four bits were determined through P_{read} and applied immediately following P_{SET} and P_{RESET} actions or after the specified delay periods. Fig. S4 (Supporting Information) presents the classification of memory states based on the conductance values read after the last read pulse in 16 different 4-bit pulse sequences. This classification demonstrates the potential of using memristors for physical storage to distinguish temporal inputs and highlights the ability of the memory states to effectively identify inputs across various time intervals. The separation of 16 memory states in RC, utilizing both P_{SET} and P_{RESET} methods, indicates that storage within the memory layer is mapped to a high-dimensional feature space. This confirms the capability of the device to capture and distinguish subtle differences in temporal inputs. Fig. 8(d) shows that, after just five epochs of training on 50,000 MNIST images, the model reached a 97.3 % accuracy in both cases, which shows the effectiveness of the described training approach. Implementing RC with both LRS and HRS modes in the W/ZnO/TiN devices has demonstrated the robustness and versatility of this approach in neuromorphic computing systems. The devices achieved high recognition rates in each scenario, which highlights their potential as effective hardware components for RC systems. In other words, the device holds significant promise for realizing hardware-based implementations of reservoir computing systems through the use of a scalable and efficient solution to process complex temporal and sequential data inputs. The capability of these devices to maintain their performance across varying resistance states further highlights their suitability for neuromorphic applications, where adaptability and dynamic learning are critical. Therefore, the outcomes of our study underscore the viability of W/ZnO/TiN as a fundamental technology that can enable the advancement of neuromorphic devices tailored for reservoir computing architectures.

4. Conclusion

Our research investigated the structural and electrical characteristics of W/ZnO/TiN RRAM devices, which confirmed their significant potential for applications in neuromorphic computing. The devices exhibited non-filamentary switching behavior and demonstrated stable durability through consistent resistance values over 500 DC cycles. Additionally, applying a read voltage decreased the current over time, and the rate of change in conductance during a 10,000 s retention measurement confirmed the STM characteristics. Furthermore, multi-level conductive characteristics were explored by applying various I_{CC} and V_{RESET} values. In addition, PPF was implemented to represent the synaptic processes of the device. Ultimately, by leveraging the STM characteristics of the device, we managed to differentiate 16 states by combining different pulse trains within an RC system. Additionally, a CNN-based readout layer was employed to process MNIST images, which resulted in high recognition rates (97.3 %) in both LRS and HRS modes.

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CRediT authorship contribution statement

Yongjin Byun: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Hyojin So:** Validation, Investigation, Formal analysis. **Sungjun Kim:** Writing – review & editing, Validation, Supervision, Project administration, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chaos.2024.115536.

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